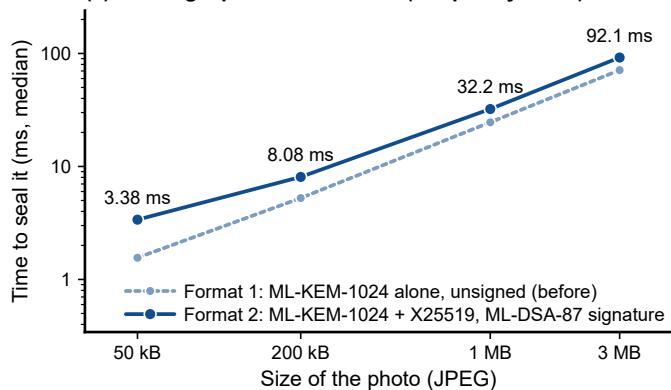
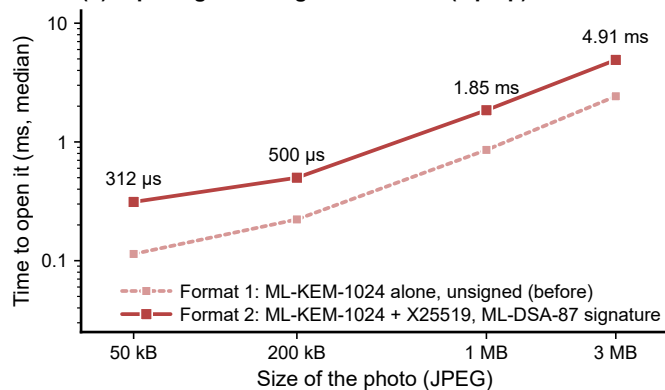
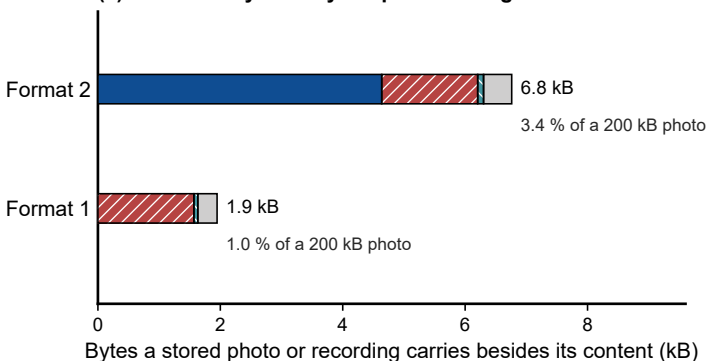
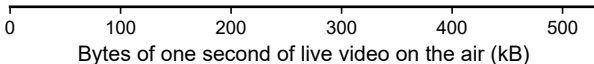


(a) Storing a photo on the UAV (Raspberry Pi 4B)**(b) Opening it at the ground station (laptop)****(c) What the hybrid key wrap and the signature add to a file**

ML-DNA-87 signature of the UAV
ML-KEM-1024 ciphertext
X25519 key, wrapped key, tag
Header (time, size, position, signer)

(d) Live video: what the AES-256-GCM record layer adds

356 records per second (11.9 per frame), each sealed and sent by the Pi in 155 μ s: 5.5 % of one of its four cores. The record layer adds 4.6 % to the video's bytes, UDP/IP another 2.7 %.



H.264 video and capture times: 375.4 kB
Record header, frame position, AES-GCM tag: 17.4 kB
UDP and IP headers: 10.0 kB